

CYBER FORENSICS CCEE CHEAT SHEET

Detailed CCEE Revision Notes

1. COMPUTER FORENSICS BASICS

Definition

Computer forensics is the scientific process of identifying, preserving, collecting, examining, analysing, documenting, and presenting digital evidence in a legally acceptable manner.

Core objectives

- Preserve evidence integrity
- Reconstruct events
- Identify attackers or users involved
- Recover deleted or hidden information
- Support legal, disciplinary, or internal action
- Present findings clearly and objectively

Main characteristics

- Repeatable
- Auditable
- Legally defensible
- Well documented
- Minimally invasive
- Integrity verified

Computer crime vs unauthorized activity

Computer Crime	Unauthorized Activity
Violates criminal law	May violate only organizational policy
May involve fraud, theft, malware, or hacking	May involve policy misuse
Can lead to prosecution	May lead to disciplinary action
Example: ransomware attack	Example: personal browsing against policy

Not every unauthorized act is a crime, but unauthorized activity may become criminal depending on intent, damage, and law.

2. CYBER LAWS

Information Technology Act, 2000

India's main legal framework for cybercrime, electronic records, digital signatures, online fraud, and unauthorized computer activity.

Important sections

Section	Meaning
Section 43	Unauthorized access, copying, disruption, virus introduction, damage
Section 65	Tampering with computer source documents
Section 66	Computer-related offences involving dishonest or fraudulent intent
Section 66C	Identity theft
Section 66D	Cheating by personation using computer resources
Section 66E	Violation of privacy
Section 67	Publishing or transmitting obscene material electronically
Section 72	Breach of confidentiality and privacy
Section 79	Intermediary liability and safe-harbour conditions

Important distinctions

- 66C: misuse of password, digital signature, or identity information

- 66D: impersonation to cheat using computer resources
 - 66E: privacy violation
 - Section 43 generally relates to unauthorized actions and compensation
 - Section 66 may apply when the act is dishonest or fraudulent
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3. FORENSIC PROCESS

Standard forensic process

Identification → Preservation → Collection → Examination → Analysis → Documentation → Presentation

Identification

Determine where evidence may exist.

Examples:

- Hard disks
- Mobile phones
- USB drives
- RAM
- Cloud accounts
- Email
- Logs
- Network devices
- Backup media

Question answered:

Where is the evidence?

Preservation

Protect evidence from alteration, destruction, contamination, or loss.

Methods:

- Use write blockers
- Seal evidence
- Photograph the scene
- Restrict access
- Calculate hashes
- Store originals securely

Question answered:

How do we keep the evidence unchanged?

Collection

Acquire the evidence using proper procedures.

Examples:

- RAM capture
- Disk imaging
- Log export
- Mobile acquisition
- Network capture

Examination

Extract useful artefacts.

Examples:

- Recover deleted files
- Extract browser history
- Analyse Registry
- Identify file signatures
- Search keywords
- Recover metadata

Question answered:

What data can be extracted?

Analysis

Interpret the extracted evidence.

Examples:

- Reconstruct timeline
- Determine attack path
- Identify exfiltration
- Correlate user actions
- Link a process to a malicious file

Question answered:

What does the evidence mean?

Documentation

Record:

- Tool name and version
- Commands
- Settings
- Dates and times
- Hashes
- Errors
- Observations
- Evidence handling

Presentation

Provide findings in:

- Final report
- Management summary
- Court testimony
- Incident review
- Technical evidence report

4. COMPUTER FORENSICS INVOLVES

Preservation

Protecting evidence from alteration.

Identification

Locating potential evidence.

Extraction

Recovering relevant data.

Documentation

Recording every action.

Interpretation

Explaining the significance of evidence.

Important distinction:

- Extraction: retrieve evidence
- Interpretation: explain evidence

5. GOALS OF FORENSIC ANALYSIS

- Identify what happened
- Determine when it happened
- Identify how it happened
- Determine who may be responsible
- Identify affected systems

- Estimate damage
- Recover deleted or hidden data
- Preserve evidence for legal use
- Prevent recurrence

Common exam phrase:

Forensic analysis is not only about finding data; it is about explaining the significance of data.

6. TYPES OF CYBER FORENSICS

Disk forensics

Analysis of storage media.

Covers:

- Deleted files
- Partitions
- File systems
- Unallocated space
- File slack
- Metadata

Memory forensics

Analysis of RAM.

Covers:

- Running processes
- Network connections
- Malware
- Encryption keys
- Loaded DLLs
- Open files

Network forensics

Analysis of network traffic and devices.

Covers:

- Packet captures
- Firewall logs
- IDS alerts
- Connections
- Protocol usage

Mobile forensics

Analysis of phones and tablets.

Covers:

- Calls
- Messages
- App data
- Location data
- Contacts
- Photos
- Cloud sync artefacts

Email forensics

Analysis of:

- Headers
- Sender paths
- Attachments
- Message IDs
- Phishing indicators

- Timestamps

Database forensics

Analysis of:

- Transactions
- Logs
- Deleted rows
- Queries
- Access history

Malware forensics

Study of:

- Malicious files
- Behaviour
- Persistence
- Network indicators
- Payloads

Cloud forensics

Analysis of:

- Audit logs
- IAM activity
- Cloud storage
- Virtual machines
- Cloud network logs

7. CYBER FORENSICS PROCEDURES

Before the incident

Preparation should include:

- Incident response plan
- Trained response team
- Contact lists
- Forensic tools
- Evidence kits
- Log retention
- Time synchronization
- Legal approval
- SOPs
- Tool validation

Incident response plan

Defines:

- Roles
- Responsibilities
- Escalation
- Communication
- Containment authority
- Evidence handling
- Reporting procedures
- Legal involvement

Incident response team

May include:

- Security analysts
- Forensic investigators
- System administrators

- Network administrators
- Legal team
- HR
- Management
- Public relations
- External vendors
- Law enforcement

Detecting incidents

Precursor

A sign that an incident may happen.

Example:

Threat intelligence reports targeting the company VPN.

Indicator

A sign that an incident may be happening or already occurred.

Example:

Multiple failed logins followed by one successful login from an unusual IP.

Important distinction:

- Precursor: future possibility
- Indicator: current or past incident

8. CHAIN OF CUSTODY

Definition

A complete documented record of every person who handled evidence and every transfer made.

Must include

- Case number
- Evidence number
- Description
- Collector
- Date and time
- Source location
- Transfer details
- Purpose of transfer
- Sender signature
- Receiver signature
- Return details
- Storage location

Importance

- Maintains accountability
- Prevents unauthorized access
- Supports admissibility
- Shows continuous control
- Detects gaps in handling

Common mistakes

- Missing signatures
- Missing transfer time
- Unrecorded checkout
- Reused seals
- Unlabelled evidence
- Evidence handled by unauthorized staff

Important distinction:

- Hash proves digital integrity

- Chain of custody proves handling accountability
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9. EVIDENCE CHECKOUT LOG

Used when evidence is temporarily removed from secure storage.

Should include:

- Evidence ID
- Person taking evidence
- Date and time removed
- Purpose
- Condition
- Date and time returned
- Receiver signature

Difference:

- Chain of custody: full handling history
 - Checkout log: temporary removal and return
-

10. HANDLING DIGITAL EVIDENCE

Basic rules

- Do not modify original evidence
- Photograph before moving
- Record serial numbers
- Use write blockers
- Seal and label properly
- Avoid unnecessary interaction
- Store securely
- Use anti-static packaging
- Document every step

Storage device packaging

Use:

- Anti-static bag
- Protective padding
- Tamper-evident seal
- Proper label

Protect from:

- Static electricity
- Shock
- Moisture
- Heat
- Magnetic fields
- Unauthorized access

Evidence labels

Should contain:

- Case number
- Item number
- Description
- Date and time
- Collector's name
- Location
- Serial number if applicable

Tamper-evident seal

Purpose:

- Shows whether packaging was opened
- Supports evidence integrity

A seal should be:

- Dated
- Initialled
- Applied across the opening

If opened:

- Record the opening
- Preserve the old seal
- Apply a new seal
- Date and initial it

11. FIRST RESPONSE

Main objective

Preserve the scene and prevent evidence loss.

First responder actions

- Secure the area
- Identify devices
- Photograph the scene
- Note system state
- Record date and time
- Document cables and connections
- Identify logged-in users
- Prevent unauthorized access
- Decide whether live collection is required

Powered-off system

- Do not boot
- Photograph
- Disconnect carefully
- Label
- Package
- Image using write blocker

Powered-on system

Consider collecting:

- RAM
- Processes
- Network connections
- Logged-in users
- Open files
- Encryption keys
- Mounted volumes
- Clipboard data

Do not immediately shut down if:

- Encryption is active
- Valuable RAM evidence exists
- Malware is running
- Remote sessions are active

Common first responder mistakes

- Rebooting the system
- Logging in normally
- Running untrusted commands
- Deleting suspicious files

- Using infected system tools
 - Failing to document actions
 - Ignoring volatile evidence
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12. RESPONSE STRATEGY

Decisions may include:

- Shutdown
- Network isolation
- Continued monitoring
- Live acquisition
- Dead-box imaging
- Immediate containment

Factors:

- Evidence preservation
- Business impact
- Threat severity
- Legal requirements
- Data sensitivity
- Criticality of system

Important principle:

There is no universal “always shut down” rule.

13. FORENSIC DUPLICATION

Bit-stream image

A complete sector-by-sector copy.

Includes:

- Active files
- Deleted data
- File slack
- Unallocated space
- File-system metadata
- Hidden partitions

Logical acquisition

Copies selected files and folders.

May exclude:

- Deleted data
- File slack
- Unallocated space
- Damaged file-system areas

Physical acquisition

Captures storage at a low level, often including all sectors.

Why create forensic images

- Preserve original evidence
- Allow repeated analysis
- Recover deleted artefacts
- Support integrity checks
- Keep master and working copies

Copy types

- Original evidence: preserved
- Master copy: protected verified reference

- Working copy: used for analysis
- Backup copy: recovery copy

14. WRITE BLOCKERS

Purpose

Prevent write commands from reaching the source drive.

Types

- Hardware write blocker
- Software write blocker

Hardware write blockers are generally preferred because they provide stronger interface-level protection.

Write blockers do not:

- Remove malware
- Decrypt data
- Reduce image size
- Recover deleted files

15. HASHING AND AUTHENTICATION

Hashing

Produces a fixed-length digest from input data.

Properties:

- Deterministic
- One-way
- Fixed-length output
- Avalanche effect
- Collision resistant
- Preimage resistant

Authentication of evidence

Shows that evidence:

- Is what it claims to be
- Has not been altered
- Was acquired properly

Supported by:

- Hashes
- Chain of custody
- Documentation
- Verified procedures

Important distinction:

- Authentication: evidence is genuine
- Attribution: linking evidence to a person

16. HASH ALGORITHMS

Algorithm	Digest Size	Hex Length
MD5	128-bit	32 characters
SHA-1	160-bit	40 characters
SHA-256	256-bit	64 characters
SHA-512	512-bit	128 characters

MD5

- 128-bit
- Known collision weakness
- Not recommended for modern security
- May still appear in older forensic tools

SHA-1

- 160-bit
- Collision weaknesses
- Better than MD5 historically, but deprecated for strong security

SHA-256

- 256-bit
- Preferred for modern integrity checks

Collision

Two different inputs produce the same hash.

Preimage resistance

Given a hash, it should be difficult to find the original input.

Second-preimage resistance

Given one input, it should be difficult to find another input with the same hash.

17. FORENSIC IMAGE FORMATS

RAW/DD

Characteristics:

- Sector-by-sector copy
- High compatibility
- Simple format
- Limited built-in metadata
- No native compression in basic form

E01

Expert Witness Format.

Supports:

- Compression
- Segmentation
- Case metadata
- Checksums
- Acquisition notes

Commonly used by FTK Imager and EnCase-compatible tools.

Ex01

Newer evidence format with metadata and improved features.

AFF

Advanced Forensic Format.

Supports:

- Metadata
- Compression
- Flexible storage

Important distinction:

- RAW/DD: simple and compatible
- E01/AFF/Ex01: metadata and compression support

18. FAILED OR DAMAGED DRIVES

For failing disks:

- Use error-tolerant imaging tools
- Minimize retries
- Log bad sectors
- Use tools designed for recovery
- Avoid unnecessary stress
- Document incomplete acquisition

A forensic image may still be usable even if some sectors are unreadable, provided:

- Errors are logged
- Limitations are documented
- Hashes are recorded appropriately

Never:

- Format the source
 - Delete files
 - Continuously retry without logging
-

19. INITIAL ASSESSMENT

Main goals

- Determine whether an incident is real
- Identify affected systems
- Determine scope
- Assess severity
- Estimate impact
- Decide escalation
- Identify immediate risks

Severity factors

- Business impact
- Data sensitivity
- Number of systems
- Criticality of asset
- Confirmed exfiltration
- Legal exposure
- Operational disruption

High-priority example:

Confirmed data theft from a critical production server.

20. INCIDENT NOTIFICATION CHECKLIST

Should include:

- Incident description
- Detection time
- Affected assets
- Severity
- Observed impact
- Current status
- Containment actions
- Contact person
- Legal or regulatory concerns
- Required escalation

Initial notification should be:

- Factual

- Concise
 - Timely
 - Action-oriented
- It should not include:

- Final attribution
- Unverified blame
- Complete technical analysis
- Speculation

21. BITS, BYTES, BINARY, AND HEXADECIMAL

Basic values

- 1 bit = 0 or 1
- 4 bits = 1 hexadecimal digit
- 8 bits = 1 byte
- 1 byte = 2 hex digits
- 8 bits represent 256 possible values
- Unsigned 8-bit range: 0 to 255

Hexadecimal digits

Hex	Decimal	Binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
A	10	1010
B	11	1011
C	12	1100
D	13	1101
E	14	1110
F	15	1111

Hex to decimal

Formula:

First digit $\times 16$ + second digit

Examples:

- $0x2F = 2 \times 16 + 15 = 47$
- $0x7C = 7 \times 16 + 12 = 124$
- $0xC5 = 12 \times 16 + 5 = 197$

Decimal to hex

Divide by 16 and convert remainder.

Examples:

- $26 = 0x1A$
- $64 = 0x40$
- $128 = 0x80$
- $255 = 0xFF$

Binary to hex

Group into 4 bits.

Examples:

- 1010 1111 = AF
- 1101 1110 = DE
- 0111 1111 = 7F

Common values

- 0x10 = 16
- 0x20 = 32
- 0x30 = 48
- 0x40 = 64
- 0x7F = 127
- 0x80 = 128
- 0xFF = 255

22. HEX EDITOR

Purpose

Displays raw bytes in:

- Hexadecimal view
- ASCII or text view
- Offset view

Uses

- Identify file type
- Inspect headers
- Inspect footers
- Locate hidden data
- Examine slack space
- Compare files
- Modify bytes on working copies
- Inspect damaged files

Offset

Position of a byte from the beginning of a file.

Example:

- 0x00 = first byte
- 0x10 = 16 bytes
- 0x20 = 32 bytes
- 0x30 = 48 bytes

If 16 bytes per row:

- Row 1 begins at 0x00
- Row 2 begins at 0x10
- Row 3 begins at 0x20

Use hex editors only on verified copies because they can modify evidence.

23. FILE SIGNATURES

File signatures or magic numbers help identify the actual file type.

File Type	Common Signature
Windows EXE/DLL	4D 5A
JPEG	FF D8 FF
PNG	89 50 4E 47
PDF	25 50 44 46

File Type	Common Signature
ZIP	50 4B 03 04
GIF	47 49 46 38
BMP	42 4D
RAR	52 61 72 21
7-Zip	37 7A BC AF
GZIP	1F 8B 08
MP3 ID3	49 44 33

Important:

- 4D 5A = MZ = Windows executable
- 25 50 44 46 = %PDF
- 50 4B = PK = ZIP-based format

ZIP-based formats may include:

- DOCX
- XLSX
- PPTX
- APK
- JAR

Trust file headers more than file extensions.

24. ENCODING VS ENCRYPTION VS HASHING

Encoding

Purpose:

- Representation
- Compatibility
- Transmission

Characteristics:

- Reversible
- No secret key required
- Not secure

Example:

Base64

Encryption

Purpose:

- Confidentiality

Characteristics:

- Reversible using a key
- Protects content

Examples:

- AES
- DES
- RSA

Hashing

Purpose:

- Integrity
- Verification

Characteristics:

- One-way
- Fixed-length digest
- No decryption

Examples:

- MD5
- SHA-1
- SHA-256

Quick comparison:

Process	Reversible	Key Needed	Main Purpose
Encoding	Yes	No	Representation
Encryption	Yes	Yes	Confidentiality
Hashing	No	No	Integrity

25. SYMMETRIC AND ASYMMETRIC ENCRYPTION

Symmetric encryption

Same key for encryption and decryption.

Examples:

- AES
- DES
- 3DES

Advantages:

- Fast
- Suitable for large data

Disadvantage:

- Key distribution problem

Asymmetric encryption

Uses public and private keys.

Examples:

- RSA
- ECC

Advantages:

- Secure key exchange
- Digital signatures

Disadvantages:

- Slower
- More computationally expensive

26. BASE64

Base64 is:

- Encoding
- Not encryption
- Reversible without a secret key
- Used to represent binary data as text

Common characters:

- A-Z
- a-z
- 0-9
- +
- /
- = for padding

Base64 usually increases data size.

27. ASCII AND UNICODE

ASCII

Maps characters to numeric values.

Examples:

- A = decimal 65 = hex 41
- a = decimal 97 = hex 61
- 0 = decimal 48 = hex 30

Unicode

Supports a much larger range of characters.

UTF-8

- Backward-compatible with ASCII
- Uses 1 to 4 bytes
- Commonly used on the web

UTF-16

- Commonly uses 2 or 4 bytes
 - Not byte-for-byte ASCII compatible
-

28. FILE CARVING

Definition

Recovering files based on headers, footers, and internal patterns instead of file-system metadata.

Useful when:

- Directory entry is deleted
- File system is damaged
- Metadata is missing
- Data exists in unallocated space

Limitations:

- Fragmented files may be recovered incorrectly
- Original filename may be lost
- Original path may be lost
- Metadata may be missing
- Encrypted data is difficult to carve

Important distinction:

- Normal recovery uses file-system metadata
 - File carving uses file content and signatures
-

29. FILE SLACK AND UNALLOCATED SPACE

File slack

Unused area between the logical end of a file and the end of its final allocated cluster.

May contain:

- Old data
- Previous file fragments
- Memory remnants

Unallocated space

Space not currently assigned to active files.

May contain:

- Deleted files
- File fragments

- Old partitions
- Data remnants

Difference:

- File slack: unused bytes within an allocated cluster
 - Unallocated space: clusters not assigned to active files
-

30. BIT ROT

Definition

Gradual degradation or corruption of stored digital data over time.

Causes:

- Aging media
- Charge leakage
- Physical damage
- Storage defects
- Environmental conditions

Detection:

- Periodic hash verification
- Read-error monitoring

Protection:

- Multiple verified copies
- Redundant storage
- Controlled environment
- Scheduled integrity checks
- Media refresh

Bit rot is usually:

- Unintentional
 - Gradual
 - Random
-

31. STANDARD OPERATING PROCEDURES

Purpose

Provide consistent, repeatable, and legally defensible methods.

SOPs should define:

- Evidence collection
- Imaging
- Tool usage
- Documentation
- Storage
- Reporting
- Quality control
- Evidence transfer
- Mobile handling
- Live response

Benefits:

- Standardization
 - Reduced errors
 - Repeatability
 - Training support
 - Legal defensibility
-

32. ACCREDITATION AND STANDARDS

General purpose

Ensure forensic laboratories and investigators follow recognized quality and competency requirements.

Important areas:

- Tool validation
- Examiner training
- Laboratory procedures
- Documentation
- Repeatability
- Peer review
- Quality assurance
- Evidence integrity

Common concepts:

- ISO/IEC 17025: testing and calibration laboratory competence
- ISO/IEC 27037: guidelines for identification, collection, acquisition, and preservation of digital evidence

Exam point:

Accreditation improves credibility but does not automatically guarantee that every conclusion is correct.

33. PRIVACY AND CYBER FORENSICS

Investigators should:

- Collect only relevant data
- Follow legal authorization
- Avoid unnecessary exposure
- Protect sensitive information
- Restrict access
- Follow retention policies
- Document scope

Privacy risks:

- Personal photos
- Medical data
- Financial records
- Private communication
- Third-party information
- Employee personal data

Important principle:

Forensic authority does not automatically mean unlimited access.

34. WINDOWS FORENSIC ARTEFACTS

Security Event Log

Useful for:

- Successful logons
- Failed logons
- Account lockouts
- Privilege use
- Audit events

Prefetch

Useful for:

- Program execution
- Last execution times
- Execution count

- File references

Important:

Prefetch availability depends on Windows version and configuration.

SAM hive

Stores:

- Local user accounts
- Password hashes
- Account identifiers

SYSTEM hive

Stores:

- System configuration
- Services
- Control sets
- Time-zone settings
- Mounted devices
- USB device information

SOFTWARE hive

Stores:

- Installed software
- Windows configuration
- Application settings

SECURITY hive

Stores:

- Local security policy
- Security-related data

NTUSER.DAT

Stores per-user Registry data:

- Recent documents
- User preferences
- Typed paths
- Application settings
- User-specific artefacts

USBSTOR

Useful for:

- USB storage device history
- Vendor and product details
- Device identifiers

Jump Lists

Useful for:

- Recently accessed files
- Application usage
- File paths

LNK files

Useful for:

- Files opened from USB drives
- Network locations
- Original path
- Volume information
- Target timestamps

pagefile.sys

Supports virtual memory.

May contain:

- Process remnants
- Credentials
- Strings
- User data
- Fragments of memory

hiberfil.sys

Contains a near-snapshot of memory from hibernation or Fast Startup.

May contain:

- Processes
- Logged-in sessions
- Encryption material
- User data

Recycle Bin

Contains:

- Deleted files
- Original paths
- Deletion timestamps
- User SID-related folders

Browser artefacts

May include:

- History
- Cookies
- Cache
- Downloads
- Autofill
- Saved sessions
- Search terms

35. WINDOWS EVENT LOG QUICK REVISION

Common Security Event IDs:

Event ID	Meaning
4624	Successful logon
4625	Failed logon
4634	Logoff
4648	Logon using explicit credentials
4672	Special privileges assigned
4688	New process created
4720	User account created
4726	User account deleted
4740	Account locked out
1102	Audit log cleared

Important:

Event logging depends on audit policy.

36. DOS AND COMMAND-LINE FORENSICS

Useful commands and artefacts:

dir

Lists files and folders.

Useful switches:

- `dir /a`: show hidden/system files
- `dir /s`: recursive listing
- `dir /t`: select timestamp field

attrib

Displays or changes file attributes.

Common attributes:

- H: hidden
- S: system
- R: read-only
- A: archive

tasklist

Displays running processes.

netstat

Displays network connections.

Useful:

- `netstat -ano`

Shows:

- Local address
- Remote address
- State
- PID

ipconfig

Displays network configuration.

Useful:

- `ipconfig /all`
- `ipconfig /displaydns`

systeminfo

Displays:

- OS details
- Boot time
- Patches
- System information

whoami

Shows current user.

net user

Displays local user accounts.

net localgroup

Displays local groups.

wevtutil

Used to query or export Windows event logs.

Important:

Running commands on a live system changes its state. Document every command.

37. SYSINTERNALS SUITE

Process Explorer

Shows:

- Running processes
- Parent-child relationships
- DLLs
- Handles
- Process properties

Process Monitor

Monitors:

- File-system activity
- Registry activity
- Process activity
- Thread activity

Autoruns

Shows:

- Startup entries
- Services
- Scheduled tasks
- Drivers
- Browser extensions
- Persistence locations

TCPView

Shows:

- TCP connections
- UDP endpoints
- Process-to-connection mapping

PsExec

Runs processes locally or remotely.

PsList

Displays process information.

PsLoggedOn

Shows locally and remotely logged-on users.

Handle

Shows open file handles.

Strings

Extracts ASCII and Unicode strings from files.

Sigcheck

Checks:

- File hashes
- Digital signatures
- Version details

Quick mapping:

- Process Explorer: processes and DLLs
- Process Monitor: real-time system activity
- Autoruns: persistence
- TCPView: network connections
- PsLoggedOn: logged-on users

38. FTK AND FTK IMAGER

FTK

Full forensic analysis suite.

Used for:

- Evidence processing
- Indexing
- Keyword search
- Email analysis
- Registry analysis
- File recovery
- Reporting

FTK Imager

Used for:

- Creating forensic images
- Previewing evidence
- Capturing memory
- Exporting files
- Hashing
- Verifying images
- Mounting images read-only

Important distinction:

- FTK Imager: acquisition and preview
- FTK: full investigation and analysis

39. OSFORENSICS

Common uses:

- File searching
- Indexing
- Deleted-file recovery
- Email analysis
- Browser history
- Password recovery features
- System information
- Hash analysis
- File signature analysis
- Case management

40. LIVE FORENSICS

Definition

Collection and analysis of artefacts from a running system.

Important live artefacts

- RAM
- Processes
- Network connections
- Open ports
- Logged-in users
- Open files
- Clipboard
- Command history

- Encryption keys
- Mounted encrypted volumes

Advantages

- Captures volatile evidence
- Accesses decrypted data
- Identifies active malware
- Shows live connections

Risks

- Alters system state
- Malware may interfere
- Tool execution creates artefacts
- Results may not be fully repeatable

Live response principle

Use trusted tools from clean external media and document:

- Command
- Time
- Tool version
- Output
- Reason

41. ORDER OF VOLATILITY

General order from most volatile to least volatile:

1. CPU registers and cache
2. RAM
3. Running processes
4. Network connections
5. Kernel and routing data
6. Temporary file systems
7. Disk data
8. Remote logs
9. Backups
10. Archival media

Exam rule:

Collect the most volatile evidence first.

42. LINUX FORENSICS

Important Linux artefacts

`/var/log`

Contains system logs.

Common files:

- `/var/log/auth.log`
- `/var/log/secure`
- `/var/log/syslog`
- `/var/log/messages`
- `/var/log/kern.log`
- `/var/log/wtmp`
- `/var/log/btmp`
- `/var/log/lastlog`

`auth.log` or `secure`

Contains:

- SSH logins
- sudo activity
- Authentication failures
- User switching

wtmp

Successful login and logout records.

Command:

last

btmp

Failed login attempts.

Command:

lastb

lastlog

Last login of each user.

Command:

lastlog

Bash history

Path:

~/.bash_history

May contain:

- Commands entered by user
- Download commands
- Privilege escalation
- File deletion

Limitations:

- Can be deleted
- May not record current session until logout
- Can be disabled

/etc/passwd

User account information.

/etc/shadow

Password hashes and aging information.

/etc/group

Group membership.

/proc

Live process and kernel information.

Useful:

- /proc/PID/cmdline
- /proc/PID/envron
- /proc/PID/maps
- /proc/net

cron

Scheduled tasks.

Locations:

- /etc/crontab
- /etc/cron.d/
- /var/spool/cron/

SSH artefacts

- ~/.ssh/authorized_keys
- ~/.ssh/known_hosts
- SSH logs

Temporary locations

- /tmp
- /var/tmp
- /dev/shm

Attackers often use them to store malicious files.

43. USEFUL LINUX COMMANDS FOR FORENSICS

ps aux

Shows running processes.

top

Shows live process activity.

netstat -antp

Shows connections and associated processes.

ss -antp

Modern alternative to netstat.

lsof

Shows open files and network sockets.

who

Shows logged-in users.

w

Shows logged-in users and activity.

last

Shows login history.

lastb

Shows failed logins.

history

Shows shell command history.

find

Searches files.

Useful:

```
find / -type f -mtime -1
```

stat

Displays file metadata.

file

Identifies actual file type using signatures.

strings

Extracts readable strings.

sha256sum

Calculates SHA-256 hash.

md5sum

Calculates MD5 hash.

dd

Creates disk images.

Example:

```
dd if=/dev/sdb of=image.dd bs=4M conv=noerror,sync
```

Meaning:

- if: input file
- of: output file
- bs: block size
- noerror: continue on read errors
- sync: pad unreadable blocks

journalctl

Queries systemd logs.

44. MOBILE FORENSICS

Main acquisition types

Manual acquisition

Examiner manually navigates and records data.

Advantages:

- Simple
- Low cost

Limitations:

- Limited
- Alters device state
- Not scalable

Logical acquisition

Collects files and data exposed by the operating system.

Examples:

- Contacts
- Messages
- App data
- Media

File-system acquisition

Collects file-system structure and more app data.

Physical acquisition

Captures raw memory or storage.

May include:

- Deleted data
- Unallocated space
- Hidden areas

May be difficult due to:

- Encryption
 - Secure boot
 - Device locks
 - Hardware restrictions
-

45. MOBILE EVIDENCE

May include:

- Calls
- SMS
- Contacts
- Photos
- Videos

- App messages
- Browser history
- Wi-Fi networks
- Bluetooth history
- GPS data
- Cloud tokens
- Email
- Social media data
- Notifications

Mobile challenges

- Encryption
- Remote wipe
- Battery drain
- Auto-lock
- Cloud dependence
- Rapid OS updates
- Proprietary hardware
- App sandboxing
- Anti-forensic features

46. MOBILE NETWORK ISOLATION

Methods:

- Airplane mode
- Faraday bag
- Shielded enclosure
- SIM removal in some cases
- Network isolation tools

Risks:

- Faraday bags may leak signals
- Device battery may drain faster
- Airplane mode interaction changes device state
- Wi-Fi or Bluetooth may remain active on some devices

Best practice:

Document every action and test isolation.

47. COMMON FORENSIC MISTAKES

- Analysing original evidence
- No write blocker
- Missing hashes
- Incomplete documentation
- Rebooting a live system
- Ignoring volatile evidence
- Unrecorded evidence transfer
- Using unvalidated tools
- Breaking seals without documentation
- Trusting file extensions
- Ignoring time zones
- Failing to record tool versions
- Modifying timestamps
- Allowing excessive access
- Mixing evidence items
- Failing to preserve original media

48. TIMELINE ANALYSIS

Important timestamp concepts

- Created time
- Modified time
- Accessed time
- Entry-modified time
- Log time
- Application time
- Network-device time

Important issues

- Time-zone differences
- UTC offset
- Clock skew
- Daylight saving time
- Manual clock changes
- Unsynchronized systems

Best practice:

Record:

- System time
 - Actual reference time
 - Time zone
 - Clock drift
-

49. CCEE HIGH-YIELD DIFFERENCES

Examination vs analysis

- Examination: extract data
- Analysis: interpret data

Hash vs encryption

- Hash: one-way, integrity
- Encryption: reversible, confidentiality

Encoding vs encryption

- Encoding: no secret key
- Encryption: requires a key

File slack vs unallocated space

- Slack: unused space inside allocated cluster
- Unallocated: not assigned to active files

Logical vs physical acquisition

- Logical: visible files
- Physical: sector-level data

Chain of custody vs evidence checkout log

- Chain: complete history
- Checkout: temporary removal

FTK vs FTK Imager

- FTK: full analysis
- FTK Imager: acquisition and preview

Prefetch vs Jump Lists vs LNK

- Prefetch: program execution

- Jump Lists: recent files and apps
- LNK: target file path and location

pagefile.sys vs hiberfil.sys

- pagefile.sys: virtual memory
- hiberfil.sys: hibernation memory snapshot

Precursor vs indicator

- Precursor: incident may occur
- Indicator: incident may be occurring or has occurred

Authentication vs attribution

- Authentication: evidence is genuine
- Attribution: links activity to a person

50. LAST-MINUTE MEMORY TABLE

Topic	Remember
First forensic priority	Preserve evidence
Volatile evidence	RAM
Disk protection	Write blocker
Exact disk copy	Bit-stream image
Integrity check	Hash
Handling history	Chain of custody
Temporary evidence removal	Checkout log
Identity theft	Section 66C
Personation fraud	Section 66D
File execution	Prefetch
User Registry	NTUSER.DAT
Local accounts	SAM
USB history	USBSTOR
Recent files	Jump Lists
Shortcut evidence	LNK
Virtual memory	pagefile.sys
Hibernation memory	hiberfil.sys
Windows process tool	Process Explorer
Persistence tool	Autoruns
Network connection tool	TCPView
Disk imaging tool	FTK Imager
4D 5A	EXE/DLL
FF D8 FF	JPEG
89 50 4E 47	PNG
25 50 44 46	PDF
50 4B 03 04	ZIP
MD5	128-bit
SHA-1	160-bit
SHA-256	256-bit
Base64	Encoding
AES	Symmetric
RSA	Asymmetric
File recovery by signature	File carving
Gradual corruption	Bit rot
Running system analysis	Live forensics
Linux login history	wtmp / last
Linux failed logins	btmp / lastb
Linux authentication log	auth.log or secure

Topic	Remember
Disk imaging command	dd

51. PRACTICAL COMMAND QUICK SHEET

Windows

```
tasklist
netstat -ano
ipconfig /all
ipconfig /displaydns
systeminfo
whoami
net user
net localgroup
wevtutil qe Security
```

Linux

```
ps aux
top
ss -antp
netstat -antp
lsof
who
w
last
lastb
lastlog
history
stat filename
file filename
strings filename
sha256sum filename
md5sum filename
journalctl
```

Disk imaging

```
dd if=/dev/sdb of=evidence.dd bs=4M conv=noerror,sync
```

Hash verification

```
sha256sum evidence.dd
md5sum evidence.dd
```

52. CCEE EXAM STRATEGY

For procedure-based questions, usually choose the option that:

- Preserves evidence
- Minimizes alteration
- Uses verified copies
- Documents actions
- Maintains chain of custody
- Uses trusted tools
- Collects volatile evidence first
- Avoids unnecessary shutdown
- Avoids analysing original media

Be careful with absolute words:

- Always
- Never
- Guarantees

- Automatically
- Only

These are often wrong unless the concept genuinely requires them.

For tool questions:

- Process Explorer: process details
- Autoruns: startup and persistence
- TCPView: network connections
- FTK Imager: image creation
- FTK: full analysis
- Hex editor: raw bytes
- USBSTOR: USB history
- Prefetch: execution
- Jump Lists: recent file use
- LNK: target file information

For hashing questions:

- Same bytes produce same hash
- Different hash usually means changed content
- Hashing does not prove ownership
- Hashing does not prove legal seizure
- Hashing does not decrypt anything

For live system questions:

- Photograph first
- Collect RAM
- Record network connections
- Record processes
- Note logged-in users
- Avoid rebooting
- Isolate network if needed
- Document every command